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August 26, 1963

SUPERSEDING
Fed. Spec. OO-M-66
December 3, 1940

FEDERAL SPECIFICATION

BREAD-SLICING MACHINE, HAND AND ELECTRIC OPERATED, SINGLE ROTATING KNIFE AND ELECTRIC-OPERATED MULTIPLE KNIFE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand-operated and electric bread-slicing machines.

1.2 Classification.

1.2.1 *Types and classes.* Bread-slicing machines furnished under this specification shall be of the following types and classes (see 6.2):

Type I.—Hand operated, single rotating knife.

Type II.—Electric operated, single rotating knife.

Type III.—Electric operated, multiple knives.

Class 1.—With gravity loaf feed chute.

Class 2.—With power loaf feed conveyor.

Class 3.—With single loaf pusher.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

CC-M-636 — Motor, (Alternating-Current), Fractional Horsepower.
CC-M-641—Motor, Alternating Current (Integral Horsepower).
QQ-N-288 — Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
QQ-N-290—Nickel Plating (Electrodeposited).
QQ-S-766—Steel Plates, Sheets, and Strip—Corrosion Resisting.
TT-P-664—Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-P-271—Paper, Wrapping, Waterproofed Kraft.
UU-T-111 — Tape, Paper, Gummed (Sealing and Securing).
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Stand-

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ards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods Of.
MIL-B-121—Barrier Material, Greaseproofed, Flexible.
MIL-A-140—Adhesive, Water-Resistant, Water Proof Barrier-Material.
MIL-B-11748—Interference Reduction for Electrical and Electronic Equipment.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. (UL) Publication:

Electrical Appliance and Utilization Equipment List.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago 11, Illinois; 550 Dundee Road, Box 247, Northbrook, Illinois; 161 Sixth Avenue, New York 13, N. Y.; or 1655 Scott Blvd., Santa Clara, California.)

Baking Industry Sanitation Standards Committee Publication:

Number 13—Bread and Roll Slicers and Wrappers.

(Application for copies should be made to Baking Sanitation Standards Committee, 511 Fifth Avenue, New York 17, New York.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a sample of the bread slicing machine shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product. The bread slicing units covered by this specification shall be the manufacturer's standard product except for any design modification and additional requirements specified herein.

3.3 Certification.

3.3.1 *Fire and safety hazards.* Prior to award of contract, the bidder shall submit to the procuring agency proof that all electrical materials or equipment he proposes to supply under this specification conforms to applicable standards of Underwriters Laboratories, Inc. (UL), regarding fire and

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casualty hazards. The label or listing with re-examination of the UL will be accepted as conforming to this requirement. In lieu of the label, or listing with re-examination, the bidder may submit independent proof satisfactory to the procurement agency, that his material or appliances conform to the published standards including method of test, of the UL. Compliance with the above preliminary requirements regarding fire and casualty hazards does not absolve the bidder from complete compliance with the other requirements of the specification in order to secure the acceptance of his material or appliance.

3.3.2 If inclusion in the "Electrical Appliance and Utilization Equipment List" or supplements thereto, is offered as evidence of compliance with the requirements established by UL, the bidder shall furnish a certificate stating that the model to be furnished is identical to the model listed in the "Electrical Appliance and Utilization List" and shall, upon demand, make available to the authorized representative necessary test data to support the certification.

3.3.3 If a certificate of a recognized independent testing laboratory is submitted as evidence of compliance with the requirements of the UL, the supplier shall furnish a certificate stating that the model offered is identical to the model tested by the independent laboratory and shall upon demand make available for examination by the contracting officer the necessary test data as developed by the independent testing laboratory to support this certification.

3.3.4 Upon demand by the contracting officer the supplier shall furnish a copy of the drawing, photographs, and other descriptive material approved by the UL or other testing agency for the purpose of identifying the model to be furnished.

3.3.5 *Sanitation.* Bread-slicing machine shall be designed and constructed in accordance with the requirements of Sanitation Standard No. 13 for Bread and Roll Slicers and Wrappers as prepared by the Baking Industry Sanitation Standards Committee, except that when a conflict or requirements may exist, this specification shall take precedence.

3.4 **Design.** Each bread slicing machine shall be designed to transversely slice any loaf of bread as large as 16 inches long by 5-1/2 inches high by 7 inches wide.

3.4.1 *Type I.* The type I machine shall be designed with one knife to cut one slice at a time. The knife shall be manually driven by means of a handle mounted on a flywheel. The loaf support and loaf feeding device shall hold the loaf stationary during the slicing stroke and shall not deform any loaf being sliced. The crumb tray shall be located below the slicing knife and shall catch the crumbs that leave the knife-guard inclosure during the slicing operation. The knife guard shall inclose the knife except where the bread passes to and from the knife, and where the crumbs fall into the crumb tray. The slice receiver shall be arranged so that the slices of bread are automatically and uniformly stacked to the setting within the range of the thickness regulator, the machine shall cut the loaf into slices of equal thickness. Each slice shall be cleanly cut, and there shall be no marked change in the shape of the loaf after being sliced. The machine shall slice at a rate of not less than 60 slices per minute when tested as specified in 4.4.1.

3.4.2 *Type II.* The type II machine shall comply with the requirements of 3.4.1, except that the knife and load-feeding device shall be motor driven and the machine shall be complete with a knife scraper and a knife sharpener. The knife scraper shall be mounted on the machine for removing all particles adhering to the surface and edge of the knife by manual movement of the device into operating position when the knife is rotating. The knife sharpener shall efficiently and properly sharpen the cutting

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edge on both sides of the knife. The machine shall slice bread at a rate of not less than 120 slices per minute when tested as specified in 4.4.1.

3.4.3 *Type III.* The type III machine shall be arranged so that the knife frames are rapidly reciprocated in opposite directions and each loaf shall be completely sliced as it moves across the knives. The bread shall not move in the direction of either knife frame. Slices shall be cleanly cut and of equal thickness, and there shall be no marked change in the shape of the loaf after being sliced. The crumb tray shall be suitable for catching all crumbs that fall from the slicer and may be a pan or drawer. The machine shall slice bread, 12 inches long and 4-5/8 inches square, at a rate of not less than 12 loaves per minute when tested as specified in 4.4.1.

3.4.3.1 *Class* 1. The type III, class 1 machine shall have a chute equipped with loaf guides fixed at such an angle or angles that when the back chute is maintained fully loaded with loaves of bread, the loaves shall move by gravity to the knives, be sliced, and move by gravity to the bread stop at the front chute.

3.4.3.2 *Class* 2. The type III, class 2 machine shall have a power-driven conveyor equipped with loaf guide that automatically delivers bread, one loaf at a time, through the machine to a discharge chute or a power-driven discharge conveyor, as specified (see 3.6.17 and 6.2).

3.4.3.3 *Class* 3. The type III, class 3 machine shall have the bread transferred from a loaf support to a slice receiver by means of a pusher. The pusher shall either move the bread across the knives or the knife frame across the bread. If the knife frame moves across the bread the knife frame shall be completely inclosed so as to prevent the operator from accidental contact with the knives.

3.5 Material. The material shall conform to the applicable specifications and requirements specified hereinafter.

3.5.1 *Nickel-copper alloy.* Nickel-copper alloy shall conform to the requirements of QQ-N-288.

3.5.2 *Corrosion-resisting steel.* Corrosion-resisting steel shall conform to the chemical composition of type 301 or 302 of QQ-S-766.

3.6 Construction. The bread-slicing machine shall be constructed as specified herein. The machine shall be complete with all accessories necessary for satisfactory operation, shall be square and in alignment, and functioning components shall not require abnormal force to operate. When tested as specified in 4.4.3, the machine shall operate as intended without vibration or objectional noise. The machine shall be arranged and finished so that it can be readily cleaned and maintained. All joints and seams shall be smooth and easily cleanable without cracks or crevices. All hollow spaces shall be filled and sealed to prevent the entrance of dust, liquids, and dirt. The bench-style machine base shall be machined and drilled for securing to a table or bench with 1/2-inch hold-down bolts. Unless otherwise specified (see 6.2) type III machine shall be furnished with a floor stand. At the option of the supplier, the crumb tray, loaf guard, slice receiver, front and back chute, and discharge chute shall be constructed of nickel or chromium-plated carbon steel, corrosion-resisting steel, or nickel-copper alloy. The crumb tray may also be constructed of nickel or chromium-plated brass or aluminum-base alloy having an electrolytically oxidized finish, and the slice receiver may be constructed also of aluminum base alloy having an electrolytically oxidized finish.

3.6.1 *Motor.* Unless otherwise specified (see 6.2), the motor for the type II and type III machines shall be of the totally inclosed type, designed to operate on a nominal 120-volt, 60-cycles, single phase, a.c. system. The motor shall have the necessary char-

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acteristics to start and operate the machine under full-load conditions without exceeding the temperature for which the motor was designed. The motor shall be equipped with external overload protection. Manual reset overload protection will be accepted if this is a common practice of the manufacturer. The motor shall conform to CC-M-636 or CC-M-641, as applicable.

3.6.2 *Switch motor.* The types II and III machine shall have a readily accessible double-pole, single-throw switch provided for starting and stopping the motor.

3.6.3 *Crumb tray.* All types of machines shall have a crumb tray attached to the machine by mechanical means which shall be readily removable for cleaning.

3.6.4 *Slice thickness regulator.* The regulator for types I and II machines shall permit adjustment of slice thickness from 1/8 inch to 11/16 inch. The regulator shall have a manually operated gage suitable for setting the desired thickness of slicers and shall include marked or numbered index lines suitable for 3/8-, 7/16-, 1/2-, and 9/16-inch settings. The regulator shall not alter its fixed setting under service conditions.

3.6.5 *Knife, slicing.* The knives shall be either heat-treated carbon steel having not less than 0.85 percent carbon, nickel or chromium plated except at the cutting edge or heat-treated corrosion-resisting steel of cutlery grade, having not less than 0.60 percent carbon and 14 percent chromium or chrome-vanadium steel, tempered and polished. One quarter inch back from the cutting edge shall have a Rockwell C hardness number not less than 54, when tested as specified in 4.4.2.

3.6.5.1 *Knife guard.* The knife guards on the types I and II machines shall be fabricated from carbon steel, corrosion-resisting steel, brass, aluminum-base alloy or cast iron. The knife guard shall inclose the knife except where the bread passes to and from the knife, and where the crumbs fall into the crumb tray. The guard shall have a solid surface, and shall be so arranged that the knife is readily accessible for replacement.

3.6.5.2 *Knife frames.* The knife frames on the type III machines shall be fabricated from carbon steel, corrosion-resisting steel, nickel-copper alloy, magnesium-base alloy, or malleable iron. The frame shall have a mechanical means for adjusting the tension of each individual knife independently. The frame shall be constructed so as not to be deformed by loads arising from the tensioning of the knives.

3.6.6 *Bearings.* Bearings shall conform to the manufacturer's standard commercial product recommended for this type of application.

3.6.7 *Gears.* Gears shall be cast iron, steel, brass or bronze. Teeth of gears shall be machine cut.

3.6.8 *Loaf support and feeding device.* The unsliced loaf support and loaf feeding device on the types I and II and type III, class 3 machines shall be carbon steel, corrosion-resisting steel, nickel-copper alloy, brass, aluminum-base alloy or iron. If carbon steel, iron, or brass is used, surfaces which contact the bread shall be nickel or chromium plated; if aluminum-base alloy is used, surfaces which contact the bread shall have an electrolytically oxidized finish. The loaf feeding device shall move the loaf after each slicing stroke, an amount predetermined by the setting of the slice-thickness regulator. The loaf feeding device shall permit slicing of the bread up to 1/2 inch from the end of the loaf.

3.6.9 *Loaf guides.* The loaf guides on the type III, classes 1 and 2 machines shall consist of two full-length members, one on each side of the chute. The position of each member shall be readily and equally adjustable. The separation between the two members

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shall be adjustable from 6 inches to 16 inches or more. Each member shall be provided with a mechanical clamping device, or devices, convenient for rapid manipulation by hand and securely locking the member in the desired position.

3.6.10 *Slice receiver.* The slice receiver on the types I and II and type III class 3 machines shall be either readily removable with the stacked slices, or arranged to be moved with the stacked slices, away from the knife by sliding along steel slides secured to the machine frame.

3.6.11 *Knife sharpener.* The knife sharpener on the type II machine shall be of the two-stone type attached to the machine, and shall be either manually adjustable to compensate for wear of the knife and stones, or it shall be so arranged that compensation for wear is automatic. When not in use the position of the sharpener shall be such that it shall not cause the accumulation of bread particles.

3.6.12 *Floor stands.* Floor stands on the type III machines, when required, shall be either carbon steel or cast iron, and of either the cabinet type or the open-box frame type. The stand shall form a rigid and substantial support for the machine, and shall have a means for fastening it to the machine. The stands shall have holes for holddown bolts at the floor level.

3.6.13 *Lubrication.* If gears or transmission chains are employed, they shall run in an oil bath in a sealed compartment filled with oil to the proper level, and there shall be means for refilling and draining and either a protected sight gage or a marked metal dip rod for readily ascertaining the proper oil level. Bearings used for rotating members, except permanent lubricated bearing shall have oil cup, grease cup, or pressure injection type grease fittings, except that ball or roller type bearings may be grease packed and sealed, and sleeve bearings may be lubricated in connection with an oil-filled housing of which they are a part. All other bearings, except permanent bearings, used for guiding or supporting the frames, shall be accessible for lubrication. All lubrication shall be so arranged that no oil or grease can come in contact with the bread.

3.6.13.1 *Tools.* If lubrication fittings of the grease-gun injection type are used, a hand-operated grease gun shall be supplied with the machine.

3.6.14 *Back and front chute (type III, class 1).* The back chute shall have a minimum length of 28 inches measured from the knives to the back edge of the chute. The front chute shall be provided with a metal loaf stop and shall have a length of at least 8 inches, measured from the knives to the loaf stop.

3.6.15 *Conveyor.* The power-driven unsliced loaf conveyor on the type III, class 2 machine shall be of the endless belt type supported on rollers mounted on bearings. The belt shall be canvas not less than 3 ply and cleated to prevent slippage of the bread loaves. Rollers shall be either steel or wood and steel core construction. Any metal parts which come in contact with the bread shall be nickel-plated carbon steel, corrosion-resisting steel, or nickel-copper alloy. The minimum length of the conveyor shall be 32 inches, measured from the knives to the back end. The conveyor shall be motor driven through gears, steel or cast-iron pulleys with V-section belts of rubber and fabric construction, or silent-type steel transmission chains with steel or cast-iron sprockets. V-section belts, if used, shall have a tension adjusting means, which permit a take-up of not less than 1/4 inch for each linear foot of belt supplied.

3.6.16 *Discharge chute.* The front discharge chute of the type III, class 2 machine shall have a minimum length of 16 inches, measured from the knives to its end, and shall be open ended with no loaf stop.

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The loaf chute shall be horizontal or set at an angle. When the chute is set at an angle it shall have a minimum horizontal length of 8 inches from the machine to the chute end.

3.6.17 *Power-driven discharge conveyor.* When specified (see 6.2), a power-driven discharge conveyor shall be furnished with the type III, class 2 in place of the discharge chute. The power-driven discharge conveyor shall comply with the requirements of 3.6.15.

3.6.18 *Pusher.* Any metal part of the pusher on the type III, class 3 machine except the knives, which come in contact with the bread, shall be nickel or chromium plated carbon steel, corrosion-resisting steel, or nickel-copper alloy. A conveniently located lever, handle or handles shall be provided for such operations as may be necessary to manually move the pusher.

3.6.19 *Radio-interference reduction.* The bread-slicing machine shall be equipped for radio-interference in accordance with MIL-I-11748, class IIId when tested as specified in 4.4.4 (see 6.4).

3.6.20 *Finish.* Exterior surfaces and parts not specified shall be neatly finished with a durable enamel in accordance with standard commercial practice or may be nickel or chromium plated, or made of nickel-copper alloy, corrosion-resisting steel, or an aluminum-base alloy having an electrolytically oxidized finish. The interior of the knife-guard surfaces shall be enameled. Exterior surfaces or parts of nickel-copper alloy and corrosion-resisting steel shall be smooth and polished. No cadmium plating shall be used on surfaces which contact the bread.

3.6.20.1 *Chromium plating.* When chromium plating is used, the plating shall be copper, nickel, and chromium deposited in successive layers of such sequence that, after buffing and polishing operations, the outermost layer is chromium not less than 0.00001 inch thick, the layer immediately below the chromium is nickel not less than 0.00060 inch thick and the total thickness of all layers is not less than 0.00150 inch.

3.6.20.2 *Nickel plating.* Nickel plating shall conform to the requirements of type DS of QQ-N-290.

3.7 **Marking for identification.**

3.7.1 *For Department of Defense.* A nameplate complying with MIL-STD-130 shall be securely fastened in a readily accessible position on the machines.

3.7.2 *For other than Department of Defense.* Unless otherwise specified (see 6.2), the manufacturer's standard nameplate will be acceptable.

3.7.3 *Motor nameplate.* All motors shall be provided with a nameplate of corrosion-resisting steel, giving the type, serial number, electrical characteristics, size, and manufacturer's name. The electrical characteristics shall include voltage, frequency, phase, amperage, and horsepower.

3.8 **Equipment manual.** The supplier shall furnish with each bread-slicing machine an equipment manual containing complete installation, operation, and maintenance instructions and an illustrated repair-parts section. The repair-parts section shall include contractor's part number, description of parts, actual manufacturer's name and part number, quantity of parts required and cost of parts (prevailing at time of contract award). The repair-parts illustration may be a drawing or picture and shall be cross referenced to the applicable part number to provide definitive identification of the illustration. Repair parts for the same or different model end items shall not have the same part number unless the parts are interchangeable. Figure, drawing, item, plate, casting, et cetera, numbers may be used for cross reference only and shall not be used in lieu of part numbers.

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3.9 Workmanship. The finished machine shall be clean, well made and conform to the quality and grade of product established by this specification.

3.9.1 *Fabrication.* All sharp burrs, slivers or splinters shall be removed.

3.9.2 *Threaded fasteners.* Threaded fasteners shall be drawn up tight.

3.9.3 *Welding and brazing (where required).* The surface of all parts after welding or brazing shall be free from burn holes, cracks, fractures, or incomplete fusion. All scale or flux deposits shall be removed from finished welds.

3.9.4 *Soldering (when applicable).* All surfaces to be soldered shall be thoroughly cleaned prior to soldering. Soldering shall be smooth, sound and clean.

3.9.5 *Electrical assembly.* Wiring shall be properly joined and shall not be cut or abraded; wiring shall not be loose at terminal end, or otherwise impaired.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for defects in accordance with 4.3.3.1 and dimensions specified, and shall be tested in accordance with 4.4.1, 4.4.3, and 4.4.4.

4.3 Inspection for acceptance. Inspection shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 *Inspection of components and materials.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards, or, if none, in accordance with this specification.

4.3.2 *Intermediate testing.* The slicing knives shall be tested as specified in 4.4.2 to determine compliance with 3.6.5. The sample unit shall be one knife. The inspection lot shall be all the knives submitted for inspection at one time. The requirement shall be applicable to the individual unit and the result shall be reported as pass or fail. There shall be three determinations made per unit and the inspection level shall be L3. If the average of the three determinations are below the minimum required, the lot shall be rejected.

4.3.3 *End-product-inspection.* All of the machines of the same type and class offered for acceptance at one time shall be considered a lot for the purpose of inspection. The sample unit for this inspection shall be one completely fabricated bread slicing machine.

4.3.3.1 *Visual examination.* The completely assembled end item shall be examined for defects in finish, construction, workmanship and marking. Defects shall be classified as outlined, as follows. The inspection level shall be level II with acceptable quality levels of 2.5 for major defects and 6.5 for total defects, expressed in defects per hundred units.

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Examine	Defect	Classification	
		Major	Minor
Finish	Type of finish not as specified or blistered, peeled, chipped, or area of no film or thin film		X
Design	Characteristic not in accordance with the specified requirements	X	
Construction	Part missing or not specified type or size	X	
	Functioning components that require abnormal force to operate		X
	Cracks and crevices not filled and sealed to prevent entrance of dust, liquid, and dirt ...	X	
	Bench style machine base not machine or drilled for specified mounting bolts	X	
	Floor stand (when required) not provided with holes for holddown bolts	X	
	Crumb tray not attached to the machine by mechanical means	X	
	Not readily removable for cleaning .	X	
	Specified components not adjustable	X	
	Components not readily accessible for servicing where required		X
Assembly	Unit perceptibly out of square or alignment	X	
Workmanship	Split, punctured, dented, deteriorated, bowed, malformed, or otherwise impaired	X	
Workmanship (cont'd)	Sharp burr, sliver, or splinter	X	
	Threaded fasteners missing, not type specified, broken, stripped, fractured, or loose		X
Welding and brazing (where required)	Not type specified, missing, incomplete, burn holes, cracked, fractured, or otherwise not fused	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed		X
Soldering (when applicable)	Missing, not adherent, or incomplete	X	
	Not clean (flux or flux residue not removed), not smooth (surface not finished neatly) or pinholes in solder		X
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal .	X	
	Adequate slack not provided for wiring to relieve strain or excessive insulating stripped from wiring		X
Tools and equipment manual	Missing or incorrect		X
Marking identification	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in the prescribed manner ..		X

4.3.3.2 *Dimensional examination.* Examination shall be made of the end item to de-

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termine compliance with dimensional requirements. Any dimension that is not within the specified requirement shall be classified a defect. The inspection level shall be L4 with an acceptable quality level of 6.5 expressed in defects per hundred units.

4.3.3.3 *Testing of the end item.* Each production unit shall be tested for operations in accordance with 4.4.3.

4.3.4 *Examination of preparation for delivery.* An examination will be made to determine that packaging, packing and marking as required by section 5 of this specification are complied with. Defects will be scored as specified below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection levels shall be L4 and the acceptable quality level shall be 4.0 defects expressed in terms of defects per hundred units.

Examination	Defect
Markings (exterior and interior container)	Missing; incorrect, illegible; of improper size, location sequence, or method of application.
Preservation	Preservative improperly applied or missing.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, crates loose, strapping, bulging, or distortion of containers.
Weight	Weight is more or less than required.

4.4 **Tests.** Tests specified in 4.4.1 and 4.4.3 for types II and III slicing machines shall be conducted with the machine connected to same electrical system specified in the invitation for bid.

4.4.1 *Functional test.* The bread slicer shall be tested with a load for ten minutes to determine that the machine slices bread at a rate specified in 3.4.1, 3.4.2, and 3.4.3. Bread for test shall be furnished by the supplier.

4.4.2 *Knife hardness test.* The knife hardness test shall conform to method 243.1 of Fed. Test Method Std. No. 151.

4.4.3 *Operational test.* Each slicer shall be operated for ten minutes without load to determine that the machine operates satisfactorily and that it complies with 3.6.

4.4.4 *Radio-interference reduction tests.* Conformance of the preproduction model or first unit of production, as applicable with MIL-I-11748 shall be determined by authorized representatives of the contracting officer using Government test equipment. The Army testing agency will furnish the contracting officer a report of the model tested, including details of the suppression system applicable for production units and recommendations for quality control of production. Upon approval by the contracting officer and provided all other requirements of this specification are met, the supplier shall examine each production unit to determine that it conforms to the report of the contracting officer and the approved suppression system of the preproduction sample or first unit of production, as applicable.

5. PREPARATION FOR DELIVERY

(For civil agency procurements, Fed. Std. No 102 should be referred to for definitions and application of the various levels of packaging and packing.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Preservation.*

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5.1.1.1.1 *Disassembly.* Projecting, detachable components shall be detached from the bread-slicing machine in order to reduce cube and prevent damage while in transit.

5.1.1.1.2 *Cleaning.* All exposed, uncoated, ferrous metal surfaces of each machine, including the slicing knives, detached parts and tools, shall be cleaned process C-1 of MIL-P-116 and thoroughly dried.

5.1.1.1.3 *Application of preservative.* Surfaces, cleaned and dried as specified, shall be coated with type P-14 preservative of MIL-P-116.

5.1.1.1.4 *Protection of preservative.* Surfaces, from which preservative might be removed by contact with packaging and packing materials, shall be covered, wrapped, or bagged with greaseproof barrier material conforming to type I or II, grade A of MIL-B-121. Barrier material shall be secured with twine made of jute, sisal, manila, or cotton, or with tape conforming to class 1 of UU-T-111, or heat sealed when heat-sealable barrier material is used.

5.1.1.1.5 *Belts and pulleys.* Uncoated ferrous metal surfaces of pulleys shall be coated with primer conforming to TT-P-664 and the primer dried. Belts shall be replaced and tension relieved.

5.1.1.1.6 *Equipment manual.* The equipment manual shall be prepared method IC-1 of MIL-P-116.

5.1.1.2 *Packaging.*

5.1.1.2.1 *Type I machine.* Each type I machine shall be packaged in a snugfitting fiberboard box conforming to type I or II, class 1, style RSC of PPP-B-636. Folding, die-cut or built-up fiberboard pads or other suitable means shall be used to immobilize the contents in such a manner as to prevent damage while in transit. Detached components, equipment manual, and tools shall be cushioned and secured in the same box as the machine. Each box shall be securely closed in accordance with the appendix of the box specification.

5.1.1.2.2 *Types II and III machines.* Components of the machine that are free to move shall be secured in such a manner as to prevent movement or vibration while in transit. Detached parts, equipment manual, and tools shall be packaged in a snugfitting fiberboard box or boxes conforming to type I or II, class 1 or 2 (see 5.2.1.2 and 5.2.2.2), style optional of PPP-B-636, respectively. Contents of each box shall be cushioned and immobilized, as necessary, with fiberboard inserts or other suitable means in such a manner as to prevent damage while in transit. Each box shall be securely closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Bread-slicing machines shall be preserved and packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.*

5.2.1.1 *Type I machine.* Type I bread-slicing machines, preserved and packaged as specified in 5.1, shall be packed in a snugfitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.1.2 *Types II and III machines.* Each type II or III bread slicing machine, preserved and packaged as specified in 5.1, shall be packed in a nailed-wood or cleated-plywood shipping container conforming to class 2, style 2 or 4, type 3 load of PPP-B-621, or overseas type, type 3 load of PPP-B-601. Each shipping container exceeding 250 pounds gross weight shall be provided with skids as specified in 5.2.1.2.1. The machine shall be secured by carriage bolts passing through the skids or through load bearing

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member attached to the inside surface of the bottom of the container, or by lumber blocking and bracing. Blocking and bracing shall not depend on end-grain nailing alone, but supplemental supporting cleats shall be provided to reinforce and prevent splitting of the blocking and bracing. The machine shall be covered with a full-height shroud made of waterproof barrier material conforming to class E-2, H-1, H-5 or M of UU-P-271. All seams of the shroud shall be sealed with a 3/4-inch minimum width continuous seam of water-resistant adhesive conforming to MIL-A-140. The fiberboard box containing detached components and tools shall be class 2 (see 5.1.1.2.2), waterproofed in accordance with the appendix of the box specification, and secured in an unused portion of the shipping container. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification.

5.2.1.2.1 *Skids.* Nailed-wood containers conforming to PPP-B-621 shall be provided with skids in accordance with the container specification. Cleated-plywood containers conforming to PPP-B-601 shall be provided with nominal 2- by 4-inch skids laid flat, positioned parallel to the ends, and extending the full exterior width of the shipping container. One skid shall be placed at each end, set in approximately three inches, and additional skids, when required, placed equidistant between the end skids. In no case shall the distance between skids exceed 48 inches. Filler cleats, of the same thickness as the container cleats and same width as skids, shall be provided between each skid and the bottom panel of the shipping container. The ends of the skids shall be beveled one-half the thickness at a 45-degree angle and notched to provide clearance for longitudinal strapping. The skids shall be secured by nailing through the bottom panel, the filler cleats, and the skids; and the nails clinched not less than one-eighth of an inch. Nails fastening the skids shall be arranged in two rows in a staggered pattern with spacing between nails in each row to be not more than six inches. Each row of nails shall be approximately one-half of an inch in from the edge of the skid. The nailing pattern shall begin and end approximately 1-1/2 inches in from the ends of each skid.

5.2.2 *Level B.*

5.2.2.1 *Type I machine.* Type I bread slicing machines, preserved and packaged as specified in 5.1, shall be packaged in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.2.2 *Types II and III machines.* Each type II or III bread slicing machine, preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.1.2 except that: Shipping containers shall conform to class 1, style 2 or 4, type 3 load of PPP-B-621, or domestic type, style A or B of PPP-B-601; fiberboard box containing detached components and tools shall be class 1; waterproofing of fiberboard box and shroud shall not be required.

5.2.3 *Level C.* Bread-slicing machines, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The bread slicing machines covered by this specification are intended for use in fixed baking installations.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of slicing machine required (1.2.1).
(*c*) Whether preproduction sample is waived (3.1).
(*d*) Motor characteristics if other than specified (3.6.1).
(*e*) If floor stands for type III slicing machine are not required (3.6).
(*f*) Whether a power driven discharge conveyor for type III, class 2 slicing machine is required (3.6.17).
(*g*) Identification nameplate, if different than manufacturer's (3.7.2).
(*h*) Selection of applicable levels of preservation, packaging, and packing (5.1 and 5.2).

6.3 Transportation description. Transportation description and minimum weight applicable to this commodity are:

Rail:

Bread-slicing machines.
Carload minimum 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Bread slicing machines.
Truckload minimum weight 24,000 pound, subject to Rule 115, National Motor Freight Classification.

6.4 Radio-interference test. The supplier shall arrange with the contracting officer for radio interference tests of the preproduction model (see 3.6.19).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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